ASTRA: Asynchronous Age-Aware Satellite Random Access via Mean-Field Control

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Motivation

Satellite IoT enables global status update services beyond terrestrial coverage, but grant-free access creates a tradeoff between freshness, energy, and congestion.

- **Challenge:** Can devices maintain fresh updates using only their local AoI, without observing other devices?
- **Coupled decisions:** Repetition and receiver diversity improve decoding but increase interference and cost.
- **Research gap:** Existing AoI-aware access models often overlook asynchronous capture-SIC and joint adaptation of repetition and resource-pool diversity.

ASTRA combines local AoI-based control with mean field modeling to account for population-induced congestion and derive age threshold policies.

System Model

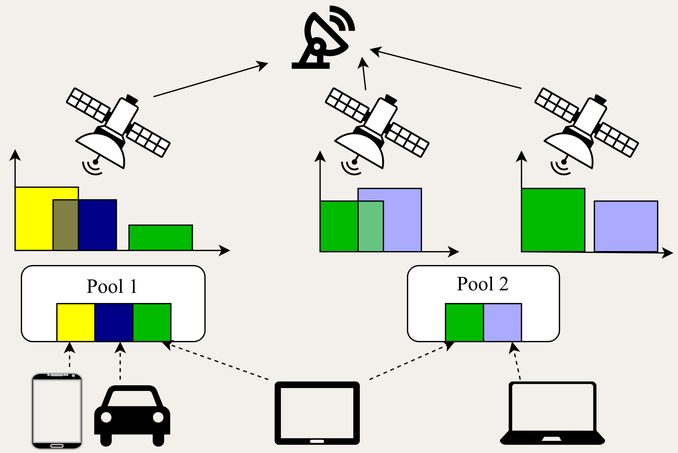


Fig. 1. System model.

We consider N devices accessing a satellite/gateway through R parallel resource pools. Each frame has duration T_f and contains M slots of duration $T_s = T_f/M$.

Access action. At the beginning of each frame, a device chooses $a = (d, q)$, where d is the number of replicas in each selected pool and q is the number of selected pools.

$$\mathcal{A} = \{(0,0)\} \cup \{(d,q) : 1 \le d \le D,\ 1 \le q \le R\}, \quad E(a) = dq.$$

Asynchronous capture-SIC physical layer. Propagation delay offsets cause partial packet overlaps, including between packets assigned to different nominal slots. Each replica experiences Rician fading, and the receiver applies SINR-based SIC. A device succeeds if at least one of its replicas is decoded in any selected pool.

Success-probability interface:

$$\widehat{p}(a;\Lambda) = \Pr(Y = 1 \mid a, \Lambda),$$

where $Y \in \{0,1\}$ is the frame-level success indicator and Λ is the per-pool replica-start intensity. The calibrated interface captures the dependence of success on both the access action and population congestion.

Mean Field MDP

Each device observes its local AoI Δ and chooses an access action a. For a fixed population load Λ, the AoI evolves as

$$\Delta_{k+1} = \begin{cases} 1, & \text{with probability } \widehat{p}(a;\Lambda), \\ \min\{\Delta_k + 1, \Delta_{\max}\}, & \text{otherwise.} \end{cases}$$

Resorting to the unconstrained Lagrangian scalarization

$$\min_{\pi\in\Pi} \bar{\Delta}(\pi) \quad \text{s.t. } \bar{E}(\pi) = B \quad \longrightarrow \quad \min_{\pi\in\Pi} \bar{\Delta}(\pi) + \eta \bar{E}(\pi).$$

The representative device minimizes the average cost with stage cost

$$c_\eta(\Delta, a) = \Delta + \eta E(a),$$

where η controls the freshness-energy tradeoff. For fixed Λ, define $p_a = \widehat{p}(a;\Lambda)$ and $V_\Delta = V_\eta(\Delta;\Lambda)$. The average cost Bellman equation is

$$\rho_\eta + V_\Delta = \min_{a\in\mathcal{A}} \left\{ \Delta + \eta E(a) + p_a V_1 + (1 - p_a) V_{\min\{\Delta+1,\,\Delta_{\max}\}} \right\}.$$

Equilibrium and Threshold Structure

Let $m(\Delta)$ denote the stationary AoI distribution. The induced per-pool load is

$$\Lambda(m, \pi) = \frac{N-1}{T_f R} \sum_{\Delta\in\mathcal{D}} m(\Delta) d(\pi(\Delta)) q(\pi(\Delta)).$$

For fixed η, the mean field equilibrium satisfies

$$\pi^\star_\eta \in \mathrm{BR}_\eta(\Lambda^\star_\eta), \quad m^\star_\eta = \mu(\pi^\star_\eta, \Lambda^\star_\eta), \quad \Lambda^\star_\eta = \Lambda(m^\star_\eta, \pi^\star_\eta),$$

where BR is the set of optimal policies for fixed Λ and multiplier η, and μ is the stationary distribution of the Markov chain induced under policy π.

Threshold structure. Define the relative value gap

$$h(\Delta) = V_\eta(\Delta + 1; \Lambda) - V_\eta(1; \Lambda).$$

The optimal action satisfies

$$a^\star(\Delta) \in \arg\min_{a\in\mathcal{A}} \left\{ \eta E(a) - \widehat{p}(a;\Lambda) h(\Delta) \right\}.$$

The optimal policy is threshold-type: conservative access at low AoI and more intensive access as AoI increases.

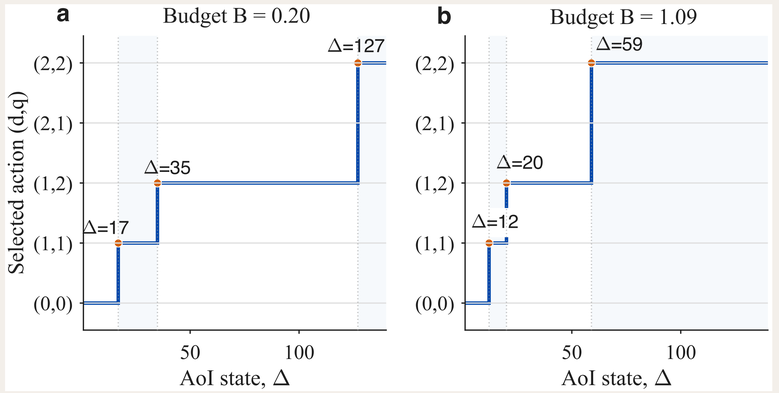


Fig. 2. Threshold structure of the optimum policy.

Numerical Evaluation

The success probability is calibrated using an asynchronous packet-level simulator with Rician fading and capture-SIC.

Baselines: IRSA-inspired access with fixed repetition distributions and optimized age-agnostic randomized access. Both are evaluated at matched average replica budgets.

Main findings:

- ASTRA achieves substantially lower average AoI.
- AoI-dependent access allocates transmission effort selectively to stale updates.
- The resulting policies exhibit the predicted age-threshold structure.

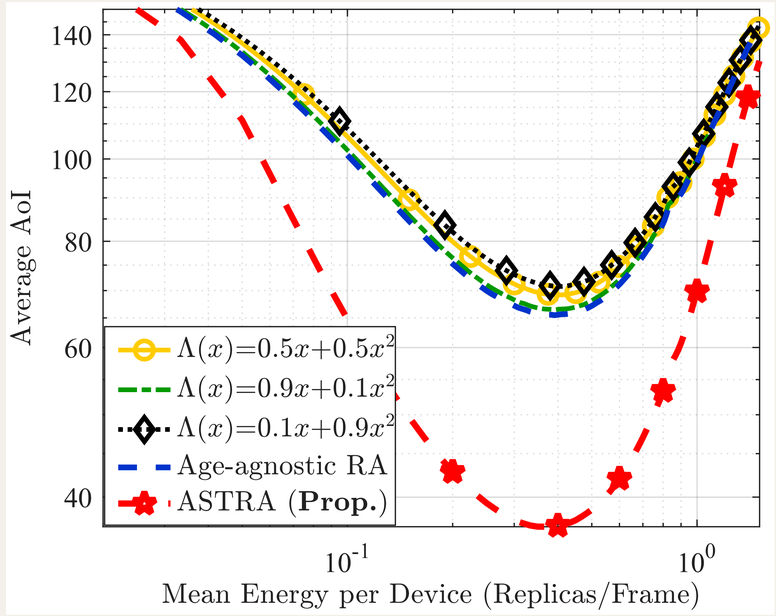


Fig. 3. Average AoI versus mean energy per device for the proposed ASTRA scheme and baseline policies.

ASTRA exploits local AoI to adapt transmission effort, achieving lower average AoI than age-independent baselines at matched replica budgets.

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